

0 4 . 1Give **two** similarities in the movement of substances by diffusion and by osmosis.**[2 marks]**

1

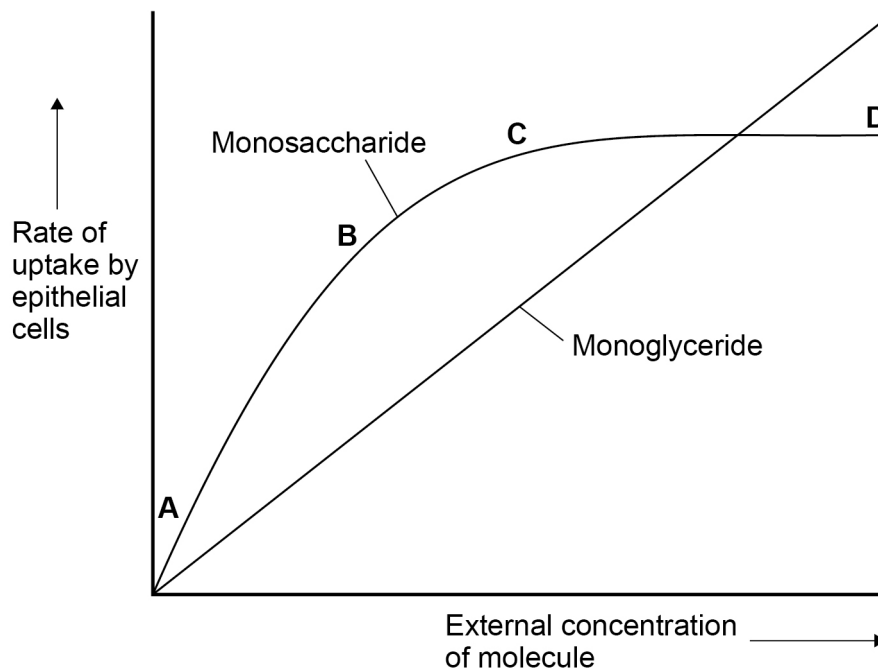
2

Question 4 continues on the next page**Turn over ►**

A scientist measured the rate of uptake of a monoglyceride and a monosaccharide by epithelial cells of the small intestine of mice. A monoglyceride is a molecule of glycerol with one fatty acid attached. She did this for different concentrations of monoglyceride and monosaccharide.

Her results are shown in **Figure 1**.

Figure 1



0 4 . 2

Use your knowledge of transport across membranes to explain the shape of the curve in **Figure 1** for uptake of monosaccharides between concentrations:

[3 marks]**A and B**

C and D

0 4 . 3

Figure 1 is evidence for monoglycerides being lipid-soluble molecules.

Suggest how.

[2 marks]

